

Companion Note for *graviton_basis_toolkit.wl*

Generalized xAct workflow for graviton operator bases, IBP reduction, and Fierz-Pauli field-redefinition reduction

Prepared as a practical guide to the script structure, inputs, outputs, and extension strategy.



This note explains what the toolkit computes sector by sector, how the caching layer is organized, how to interpret the quotient bases, and how to recover the field redefinitions that remove redundant operators.

1. Sector organization and counting conventions

The script works with scalar operator sectors labeled by the number of graviton fields n_h and the total number of derivatives N_∂ . For a linearized graviton $h_{\mu\nu}$ in four dimensions, the engineering mass dimension is taken to be 1, so the total operator dimension is

$$D = n_h + N_\partial.$$

At fixed (n_h, N_∂) the code generates all scalar contractions of products of differentiated fields, canonically simplifies them with xAct, and then performs two quotient operations:

Stage	What it removes	How it is implemented
Raw basis	nothing	AllContractions on differentiated h tensors, followed by canonicalization
IBP quotient	total derivatives	vanishing Euler-Lagrange derivative of a general linear combination
FP quotient	terms removable by field redefinition images induced by $h \rightarrow h + \Delta h$ inside the quadratic Fierz-Pauli action	

For the cubic two-derivative sector, this reproduces the workflow already tested in your notebook: start from the 16 raw $h(\partial h)(\partial h)$ structures, quotient by the two IBP relations, and then quotient further by the Fierz-Pauli redefinition image.

2. IBP algorithm

Suppose a raw sector basis is $O_i, i = 1, \dots, N$. The script forms an ansatz

$$L_{\text{ans}} = \sum c_i O_i,$$

and computes the Euler-Lagrange derivative with respect to $h_{\mu\nu}$. A linear combination is IBP-trivial precisely when its Euler-Lagrange derivative vanishes identically. Therefore the linear system

$$\delta L_{\text{ans}} / \delta h_{\mu\nu} = 0$$

determines all IBP relations among the raw operators. In the code this is done with `VarD[h[-m, -n], PD]`. The solution space is converted into a relation matrix, pivot columns are identified, and a quotient basis is built by deleting a pivot representative from each relation class.

3. Fierz-Pauli field-redefinition quotient

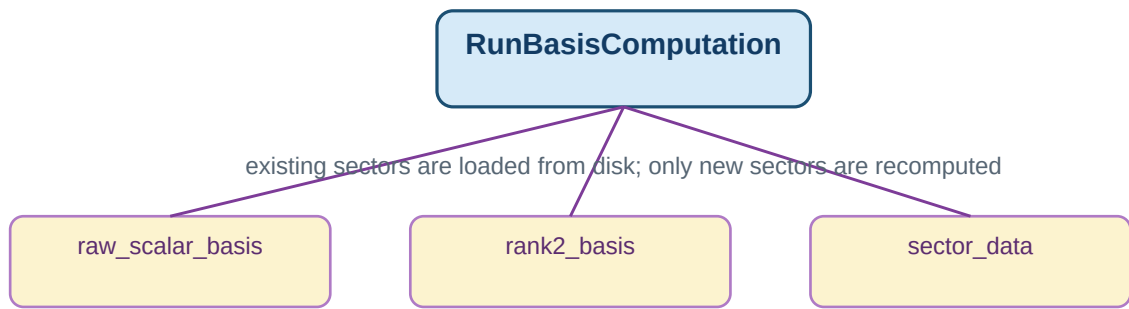
Let the free quadratic action be the Fierz-Pauli Lagrangian L_{FP} . A local field redefinition

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + \lambda \Delta h_{\mu\nu}$$

changes the action by the first-order shift

$$\Delta L = (d/d\lambda) L_{\text{FP}}[h + \lambda \Delta h] |_{\lambda=0}.$$

For a target scalar sector (n_h, N_∂) , the relevant redefinitions have the schematic form $\Delta h \sim (\partial)^{N_\partial-2} h^{n_h-1}$. The script constructs a basis of such symmetric rank-2 tensors with free indices, injects them into the shift of L_{FP} , and projects the resulting scalar images onto the IBP quotient basis. The span of those projected vectors is the field-redefinition image. Removing its pivot representatives yields the final physical basis.



4. Script structure

Setup

Defines the manifold, flat metric, graviton tensor h , and auxiliary tensors dh and $probe$.

RawScalarBasis

Builds all scalar contractions in a sector by distributing derivatives among fields and canonicalizing the contractions.

IBPData

Computes the IBP relations by the Euler-Lagrange criterion and returns the relation matrix plus quotient-basis data.

Rank2Basis

Builds the redefinition tensor basis by first generating probe-linear scalars and then varying with respect to the symmetric probe tensor.

ShiftFromRedefExpr

Computes the cubic or higher-order scalar image induced by a given Δh insertion into the quadratic FP action.

RedefinitionImageData

Projects all redefinition images into the IBP quotient basis and extracts an independent spanning set.

ComputeSectorData / RunBasisComputation

Assembles sector summaries and loops over all sectors up to the requested maximum dimension.

5. Core usage pattern

```
res = RunBasisComputation[
  "MaxDimension" -> 6,
  "CacheDirectory" -> FileNameJoin[{NotebookDirectory[], "basis_cache"}],
  "UseCache" -> True,
  "ForceRecompute" -> False,
  "Verbose" -> True,
  "ExportSummaryCSV" -> True
];

PrintSectorSummary[res];

phys32 = PhysicalBasisOfSector[res, 3, 2];
redefs32 = RedefinitionRulesOfSector[res, 3, 2];
```

6. Main outputs to inspect

Accessor	Meaning
RawBasisOfSector[res, nh, Nd]	All canonical scalar contractions before quotienting
IBPBasisOfSector[res, nh, Nd]	Basis after quotienting by total derivatives
PhysicalBasisOfSector[res, nh, Nd]	Basis after both IBP and FP field redefinitions
RedefinitionsOfSector[res, nh, Nd]	Independent Δh tensors whose FP images span the removable subspace
RedefinitionRulesOfSector[res, nh, Nd]	Coefficient data expressing removable directions in the IBP basis

7. Caching behavior

Caching is file-based and sector-local. Intermediate objects such as the raw scalar basis, the rank-2 redefinition basis, and the assembled sector data are stored under deterministic filenames. If you first run up to $D = 5$ and later rerun with $D = 6$ using the same cache directory, all pre-existing sectors are loaded immediately and only the new $D = 6$ sectors are computed. This is especially useful because the $D = 6$ run naturally revisits lower-dimensional sectors unless those cached objects are available.

8. Extending to multi-field systems

The same architecture extends to an EFT with several fields ϕ^A in addition to $h_{\mu\nu}$. The key change is that a sector must now record field content as a multidegree, for example $(n_h, n_{\phi}^{\mu\nu}, n_{\chi}, \dots; N_{\partial})$. The raw-basis generator then allows differentiated factors from several tensor species, while the IBP test is still performed on the scalar ansatz exactly as before.

For field redefinitions, one must compute the image of the free equations of motion of each field. Schematically, if the free quadratic action is $S_0[\Phi^I]$, a general local redefinition $\Phi^I \rightarrow \Phi^I + \Delta\Phi^I$ induces $\Delta L = \sum_I E_I \Delta\Phi^I + \text{total derivative}$,

where $E_I = \delta L_0 / \delta \Phi^I$ are the free equations of motion. In practice this means building a redefinition basis for each field species, computing its image under the free quadratic action, and combining all projected image vectors before quotienting the sector basis.

9. Practical caveats

- Validate the first nontrivial sectors on your installation before launching a long run.
- For large dimensions, raw contraction generation can grow rapidly; use SectorList to target only sectors needed in the paper.
- Keep derivative order and field content explicit in cache keys so that changing conventions does not silently reuse incompatible files.
- When interpreting the final basis physically, remember that the script removes redundancies relative to the chosen free quadratic action and the chosen field content.

10. Recommended workflow

First run the script up to $D = 5$ and confirm that the cubic two-derivative sector matches your hand analysis. Then turn on caching and grow dimension one layer at a time. When a sector is important for the paper, export its physical basis and independent redefinitions explicitly so that the analytic argument in the text and the computational appendix can be checked side by side.

Associated script: graviton_basis_toolkit.wl